SOP Number and version: SOP-NAT-CFU.v01

SOP Title: **Bacteria Enumeration**

Table 1. SOP current version review and approval table

	NAME	TITLE	SIGNATURE	DATE
Author	Research name	Student		01/10/2024
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Approval	Frank J. Hernandez	Principal investigator		09/01/2025

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1. Overview and background

Enumeration of bacteria provides the information on the number of bacteria in a sample. This is critically important for preparing certain number of bacteria for an experiment or understanding the effect of an infection/treatment in-vivo.

2. Scope

In this SOP, bacteria enumeration methods (Bacteria counts) used in Nucleic Acids Technology Laboratories located at Linköping University (Campus Valla, B-huset, Rum 2E:686) will be explained.

This SOP covers the sections below.

- Sample preparation
- Enumeration

3. Procedure

3.1. Sample preparation

3.1.1. Liquid samples/cultures

Liquid samples/cultures do not require prior homogenization, but vortexing is important to make sure that the sample is homogenous.

- Prepare 7 Eppendorf (D1-D7) tubes with 900 µL PBS.

- Vortex the sample until it is homogenous and transfer 1 mL into an Eppendorf tube (D0: non-diluted sample).
- Take 100 µL from D0 and add this into the first tube (D1: 10^1 times diluted) and vortex.
- Perform the serial dilutions **(Figure 1)** by transferring 100 µL until the last tube (D7: 10^7 times diluted).
- Take 10 µL from each tube (D0-D7) and inoculate onto an agar plate with a non-selective media. Inoculate 3 plates from each sample.
- Use semi-quantitative continuous streak method as described in **SOP-NAT-CULT**.
- Incubate the plates at 36±1°C for 18-24 hours.
- Count the number of colonies in each plate using an automated colony counter **(Figure 2).**
- Take pictures of each plate counted.
- Calculate the average of the three plates for each dilution.
- Multiply the average number of colonies with dilution coefficients (1 to 10^7) for each dilution.
- Multiply these numbers with 100 (inoculation coefficient) to reach the number of bacteria in 1 mL (since inoculation volume is 10 µL).
- Use the colony numbers in different dilutions to increase the sample size and accuracy of the enumeration.
- Report the results as CFU/mL using the format described in **SOP-NAT-DATA**.
- If the number of bacteria in the sample/culture is predicted, use only the related dilutions to save time and materials (e.g., if 10^8 CFU/mL is expected, use only D4-D5-D6-D7 for the enumeration since the lower dilutions will contain too many bacteria colonies to count)

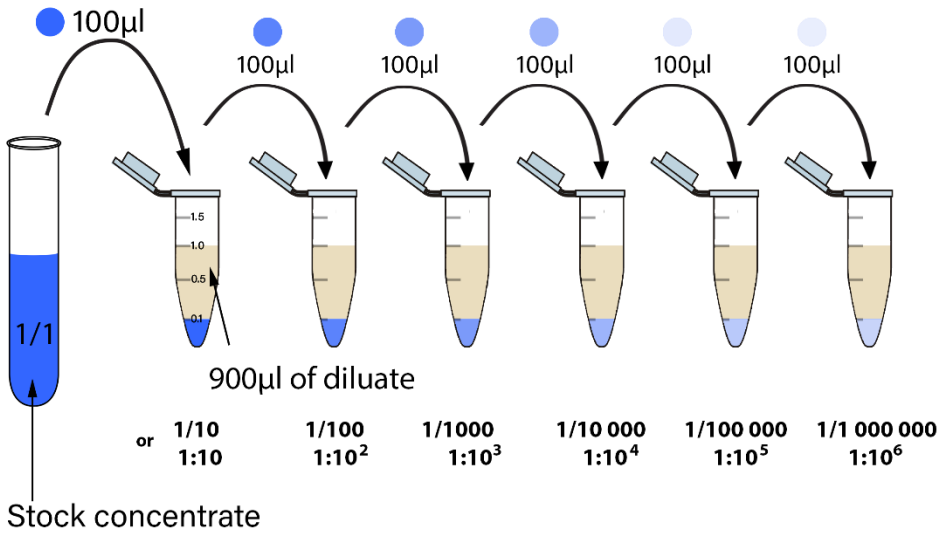

Figure 1. Example of serial dilutions of a liquid sample for bacteria enumeration.

Figure 2. Automated colony counter

3.1.2. Solid samples (e.g., tissue biopsy)

Solid samples need to be homogenized before enumeration for obtaining accurate results.

- Weigh the sample using high precision analytical lab scale.
- Transfer the sample to the screwed capped homogenization tubes **(Figure 3**) with beads according to recommendations of the manufacturer (e.g., ceramic beads for muscle tissue).
- Add 1 mL buffer (e.g., PBS) and place the tubes into the automated tissue homogenizer **(Figure 4)** compatible with the homogenization tubes used.
- Homogenize the sample at highest speed for 5 minutes (This may need to be optimized for each sample type).
- Make sure that the homogenization occurred by visually inspecting the content.
- Take 100 µL from the homogenization tube (D0) and add this into the first tube (D1: 10^1 times diluted) and vortex.
- Perform the serial dilutions **(Figure 1)** by transferring 100 µL until the last tube (D7: 10^7 times diluted).
- Take 10 µL from each tube (D0-D7) and inoculate onto an agar plate with a non-selective media. Inoculate 3 plates from each sample.
- Use semi-quantitative continuous streak method as described in **SOP-NAT-CULT**.
- Incubate the plates at 36±1°C for 18-24 hours.
- Count the number of colonies in each plate using a colony counter **(Figure 2).**
- Take pictures of each plate counted.
- Calculate the average of the three plates for each dilution.
- Multiply the average number of colonies with dilution coefficients (1 to 10^7) for each dilution.
- Multiply these numbers with 100 (inoculation coefficient) to reach the number of bacteria in 1 mL (since inoculation volume is 10 µL).
- Use the colony numbers in different dilutions to increase the sample size and accuracy of the enumeration.
- Divide these results (CFU/mL) by the tissue weight to obtain CFU/gram tissue.
- Report the results using the format described in **SOP-NAT-DATA**.

- If the number of bacteria in the sample is predicted, use only the related dilutions to save time and materials (e.g., if 10^8 CFU/mL is expected, use only D4-D5-D6-D7 for the enumeration since the lower dilutions will contain too many bacteria colonies to count).

Figure 3. Homogenization tubes.

Figure 4. Automated tissue homogenizer

4. Purpose/Objective

The objective of this SOP is to guide Nucleic Acids Technology staff on bacteria enumeration practices in Nucleic Acids Technology Laboratories.

5. Definition / Explanation of key terms

- SOP – Standard Operating Procedure
- R&D – Research and Development
- CTO – Chief Technology Officer
- CSO – Chief Scientific Officer
- COO – Chief Operations Officer
- PPE – Personal Protective Equipment
- µL – Microliter
- mL – Milliliter
- CFU – Colony Forming Units
- PBS – Phosphate Buffered Saline

6. Responsibilities

This SOP was prepared by Student name, revised by Baris Ata Borsa and approved by Frank J. Hernandez, principal investigator.

7. Equipment/Material

- High precision analytical balance
- Glass bottles
- Ultrapure water system
- Autoclave
- Autoclave gloves
- BSL-2 biosafety cabinet

- Petri dishes
- Glass flasks
- Aluminum foil
- Inoculation loops
- Bunsen burner
- Gloves
- Pipettes
- Pipette tips
- Incubator
- Automated tissue homogenizer
- Homogenization tubes
- Automated colony counter

8. Caution/Warning

- Use BSL-2 biosafety cabinet for all the work that involves bacteria.
- Use PPE according to **SOP-NAT-BSL2**.
- Clean the working area before and after work.

9. References

- Bennett JE, Dolin R, Blaser MJ. 2014. Mandell, Douglas, and Bennett's principles and practice of infectious diseases, 8th ed, Elsevier Saunders, Philadelphia, PA.
- Versalovic J, Carroll K, Funke G, Jorgensen J, Landry M, Warnock D. 2011. Manual of clinical microbiology, 10th ed, ASM Press, Washington, DC.

10. Version Control

Original version of the **SOP-NAT-CFU.v01**. Check **Table 1**.

11.Attachments/Appendices

Appendix I:

- Bunsen burner manual
- Automated homogenizer manual
- Automated colony counter manual
- Lab scale manual
- BSL-2 biosafety cabinet manual